AP PRECALCULUS

Unit 1 • Lesson 1.3 practice • Answer key

Lesson 1.3 • Rates of Change in Linear and Quadratic Functions

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Each list gives the outputs of a function at $x = 0, 1, 2, 3$. Which list could come from a linear function?

(A) $3, 5, 9, 15$

(B) $4, 7, 10, 13$

(C) $1, 2, 4, 8$

(D) $2, 3, 5, 8$

Solution

Answer: (B). The differences are all 3: a constant rate of change.

Question 2

NO CALCULATOR

For a quadratic $f(x) = ax^2 + bx + c$, the second differences of $f(0), f(1), f(2), \ldots$ are all -4. What is a?

(A) -4

(B) 4

(C) -2

(D) 2

Solution

Answer: (C). With input step 1, the second difference equals $2a$, so $2a = -4$ and $a = -2$.

Question 3

NO CALCULATOR

Let $f(x) = -(x-2)^2 + 5$. What is the average rate of change of f on $[2, 4]$?

(A) 2

(B) 0

(C) -4

(D) -2

Solution

Answer: (D). $f(2) = 5$, $f(4) = 1$, so $\dfrac{1-5}{2} = -2$.

Question 4

NO CALCULATOR

At what value of x does the rate of change of $f(x) = 3x^2 - 12x + 1$ change from negative to positive?

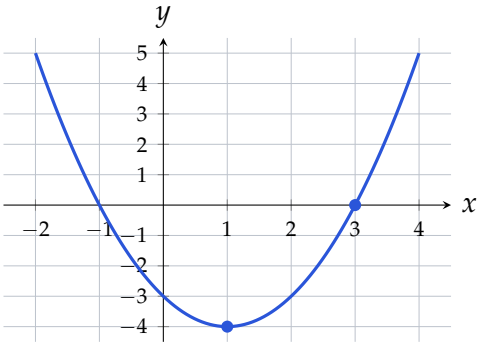
(A) $x = 2$
(B) $x = -2$
(C) $x = 4$
(D) $x = 12$

Solution
Answer: (A). The vertex is at $x = -\frac{b}{2a} = \frac{12}{6} = 2$; $a > 0$, so f decreases before it and increases after it.

Question 5 (graph)

NO CALCULATOR

The graph of a quadratic function f is shown. Its vertex is $(1, -4)$ and its zeros are -1 and 3. What is the average rate of change of f on $[1, 3]$?



(A) -2
(B) 4
(C) 2
(D) 0

Solution
Answer: (C). $f(1) = -4$ and $f(3) = 0$, so $\dfrac{0 - (-4)}{3 - 1} = 2$. On $[-1, 1]$ it is -2: the rate of change increases, as for any quadratic with $a > 0$.

Question 6 (Free response)

NO CALCULATOR

x	0	1	2	3	4
y	5	2	1	2	5

(a) Show that a quadratic function models these data exactly.
(b) Write an equation for this quadratic function.
(c) Find the average rates of change on $[0, 1]$, $[1, 2]$ and $[2, 3]$, and explain what they show about the concavity of the graph.

Solution
(a) First differences: $-3, -1, 1, 3$. Second differences: $2, 2, 2$, constant, so the data are quadratic.
(b) The vertex is $(2, 1)$ and $2a = 2$ gives $a = 1$: $y = (x - 2)^2 + 1$.
(c) -3, -1, 1. The average rates of change over consecutive equal intervals increase, so the graph is concave up.